

Sebastian K. Müller

Education

- 2015-2018 Max Planck Institute for Meteorology, Hamburg
- Ph.D. Earth System Modeling
Dissertation: *Convectively Generated Gravity Waves and Convective Aggregation in Numerical Models of Tropical Dynamics*
Adviser: Elisa Manzini
- 2007-2014 Ludwig-Maximilians-University, Munich
- M.Sc. Meteorology
Thesis: *Tornadic Supercell Thunderstorms in landfalling Tropical Cyclones*
Adviser: Roger K. Smith
- B.Sc. Physics plus Meteorology
Thesis: *Penumbra Functions for Solar Radiation Measurement Instruments*
Adviser: Stefan Wilbert
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Work Experience

- PostDoc 03/2023-03/2026 University of Trento
- Project: next generation Earth System Models
- Supervisor : Simona Bordoni
- PostDoc 09/2019-08/2022 ICTP, Earth System Physics, Trieste
- Heavy precipitation events in convection-permitting regional climate models
 - Tracking Algorithms and Inter-comparison
 - Model data postprocessing and model testing
- PostDoc 11/2018-04/2019 Max Planck Institute for Meteorology, Hamburg
- Convective Aggregation in Sea Surface Temperature Gradients
 - Contribution of Simulations to the *Radiative Convective Equilibrium Model Intercomparison Project*
 - Tutoring exercises of Prof. Bjorn Stevens' lecture *All about Atmospheric Waves*

Guest Researcher	03/2017	University of Tokyo	- Analysis of NICAM-ICSOM ₁ model output
Student Assistant	03/2014-03/2015	Meteorological Institute Munich	Tropical Meteorology working group - initialization in thermal wind balance for a numerical model - visualizations of satellite and numerical model data
Student Assistant	2009-2012	Meteorological Institute Munich	- remote sensing (LIDAR) - numerical parameterizations of radiation
Internship	08-11/2011	Potsdam Institute for Climate Impact Research	Transdisciplinary Concepts and Methods - Research Domain IV
Internship and B.Sc.-Thesis	04-10/2010	German Aerospace Center	Institute of Solar Research at Plataforma Solar de Almeria, Spain

Key Publications

Segura, H., Müller, S. K., et al. (2025). *nextGEMS: entering the era of kilometer-scale Earth system modeling*. Geoscientific Model Development 18.20 (2025): 7735-7761.

Müller, S. K., Pichelli, E., Coppola, E., et al. (2023). *The climate change response of alpine-mediterranean heavy precipitation events*. Climate Dynamics.

Müller, S. K., Caillaud, C., Chan, S., de Vries, H., et al. (2022). *Evaluation of alpine-mediterranean precipitation events in convection-permitting regional climate models using a set of tracking algorithms*. Climate Dynamics.

Müller, S. K., & Hohenegger, C. (2020). *Self-aggregation of convection in spatially varying sea surface temperatures*. Journal of Advances in Modeling Earth Systems, 12(1), e2019MS001698.

Müller, S. K., Manzini, E., Giorgetta, M. A., Sato, K., & Nasuno, T. (2018). *Convectively generated gravity waves in high resolution models of tropical dynamics*. Journal of Advances in Modeling Earth Systems, 10, 2564-2588.

Co-Authorships

Coppola, E., Raffaele, F., Giorgi, F., Giuliani, G., Xuejie, G., Ciarlo, J. M., Müller, S. K., ... & Rechid, D. (2021). Climate hazard indices projections based on CORDEX-CORE, CMIP5 and CMIP6 ensemble. Climate Dynamics, 57(5), 1293-1383.

Wing, A. A., Stauffer, C. L., Becker, T., Reed, K. A., Ahn, M. S., Arnold, N. P., Müller, S. K., . . . & Zhao, M. (2020). *Clouds and convective self-aggregation in a multimodel ensemble of radiative-convective equilibrium simulations*. *Journal of advances in modeling earth systems*, 12(9), e2020MS002138.

Smith, R. K., Montgomery, M. T., Kilroy, G., Tang, S., and Müller, S. K. (2015): *Tropical low formation during the Australian monsoon: The events of January 2013*, *Aust. Meteorol. Oceanogr. J.*, 65, 318-341.

Douillet, G. A., Taisne, B., Tsang-Hin-Sun, È, Müller, S. K., Kueppers, U., and Dingwell, D. B.: *Syn-eruptive, soft-sediment deformation of deposits from dilute pyroclastic density current: triggers from granular shear, dynamic pore pressure, ballistic impacts and shock waves*, *Solid Earth*, 6, 553-572, 2015.

Conference Contributions

<i>Presentation</i>	EGU General Assembly Conference, 2022: The Climate Response of Heavy Precipitation Events over the Alps and in the Mediterranean.
<i>PICO</i>	EGU General Assembly Conference, 2021: The Climate Response of Heavy Precipitation Events over the Alps and in the Mediterranean.
<i>Poster</i>	EGU General Assembly Conference, 2018: Sea Surface Temperature Gradients and Convective Aggregation
<i>Poster</i>	AGU Fall meeting, 2016: Tropical Convection and Gravity Waves in Spherical Limited Area Models

Teaching Experience

<i>Tutor</i>	10/2018- 04/2019	University of Hamburg <i>All About Atmospheric Waves</i> Lecturer: Bjorn Stevens
<i>Tutor</i>	05/2016	International Max Planck Research School for Earth System Modeling, MPI, Hamburg <i>Introductory Course</i>

Programming Skills

<i>Advanced</i>	python, L ^A T _E X, Unix/Linux
<i>Basic</i>	Fortran, MATLAB, git

Languages

<i>German</i>	Mothertongue
<i>English</i>	Professional
<i>Italian</i>	Fluent
<i>Spanish</i>	Intermediate
<i>French and Latin</i>	Basic

April 7, 2026